

12. Aldosterone and Mineralocorticoid Receptors_ Renal and Extrarenal Roles · assets 1-9 / 9

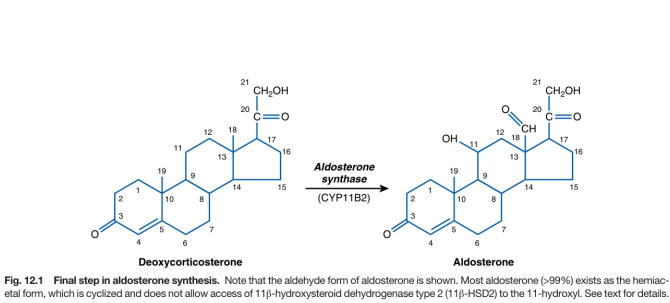


Fig. 12.1 Final step in aldosterone synthesis. Note that the aldehyde form of aldosterone (>99%) exists as the hemiacetal form, which is cyclized and does not allow access of 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2) to the 11-hydroxyl. See text for details.

Fig_12_1

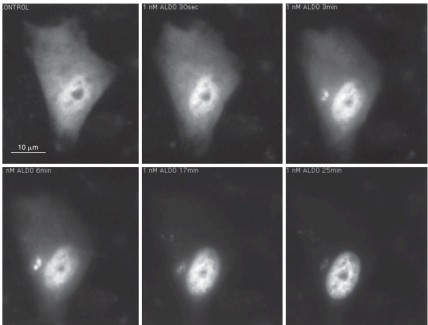


Fig. 12.4 Time-dependent nuclear translocation of the mineralocorticoid receptor (MR) in the presence of aldosterone. Cultured cells expressing green fluorescent protein-MR fusion protein (GFP-MR) were grown in a steroid-free medium and treated with 1 nM aldosterone. Translocation of GFP-MR was followed in real time, and images were captured at indicated times. It is notable that the nuclear accumulation of GFP-MR started within 30 seconds, was half-maximal at 7.5 minutes, and was complete by 10 minutes. Control MR distribution before addition of aldosterone. (From Pages Toth G, Pearce D, Narey-Pages Toth A. Subcellular localization of mineralocorticoid receptors in living cells: effects of receptor agonists and antagonists. *Proc Natl Acad Sci U S A*. 1996; 93(27):10371-10376.)

Fig_12_4

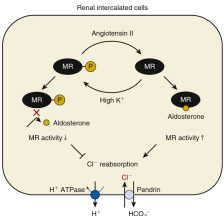
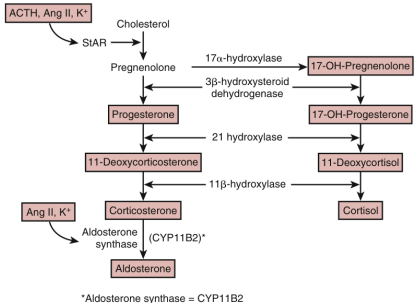


Fig. 12.7 The role of mineralocorticoid receptor (MR) ligand-hormone-binding domain (LBD) phosphorylation in controlling chloride reabsorption by intercalated cells. When phosphorylated at Ser-843 in the LBD, MR cannot bind ligands and hence cannot be activated. This phosphorylated state of MR is found only in intercalated cells, not in neighboring principal cells. In states of volume depletion, an elevated angiotensin II level decreases MR phosphorylation at Ser-843 (right side), allowing activation. In intercalated cells, MR mediates stimulation of both the proton pump (H⁺-ATPase) and Cl⁻/HCO₃⁻ exchangers such as pendrin, thereby increasing Cl⁻ reabsorption and promoting increased plasma volume while inhibiting K⁺ secretion. In contrast, in states of hypokalemia, phosphorylation of Ser-843 is increased (left side) and hence Cl⁻ reabsorption by intercalated cells is decreased and the principal cell-dependent K⁺ secretion is increased. (From Shibata S, Finckh J, Zhang J, et al. Regulated mineralocorticoid receptor phosphorylation controls ligand binding and renal response to volume depletion and hypokalemia. *Cell Metab*. 2013;18:660-671.)

Fig_12_7



*Aldosterone synthase = CYP11B2

Fig. 12.2 Overview of aldosterone synthetic pathway showing key regulatory nodes. Note that adrenocorticotropic hormone (ACTH), angiotensin II (Ang II), and K⁺ regulate steroidogenic acute regulatory protein (StAR), which stimulates cholesterol uptake by mitochondria and thus substrate availability for synthesis of all of the steroid hormones. Aldosterone synthase (gene name CYP11B2), which is selectively expressed in the adrenal glomerulosa, mediates the final step in aldosterone synthesis. It is also regulated by Ang II and K⁺. Aldosterone synthesis is shown (left). Cortisol synthesis is also shown (right) to emphasize the interconnections and similarities between these pathways.

Fig_12_2

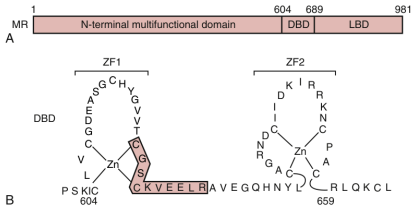


Fig. 12.5 Domain structure of the mineralocorticoid receptor (MR). The size and amino acid designations used here are for rat MR (981 amino acids total); they apply with minor variations to human MR (984 amino acids total). (A) Strip diagram of MR. The N-terminus is to the left, C-terminus to the right. (B) Schematic of MR DNA-binding domain (DBD), showing the two zinc fingers and the positions of the coordinating Zn ions. Boxed region is the α helix, which intercalates into the major groove of the DNA and provides the major protein-DNA interaction contacts. The dimerization interface comprises amino acids within the second zinc finger, which form van der Waals and salt bridge interactions. LBD, Ligand/hormone-binding domain.

Fig_12_5

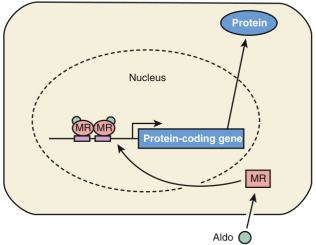


Fig. 12.3 General mechanism of aldosterone action through the mineralocorticoid receptor (MR). This simple schematic shows the general features of MR regulation of a "simple" hormone response element (HRE), common to a large subset (but not all) of aldosterone-stimulated genes. Note that in the absence of hormone, MR is found in both nucleus and cytoplasm (see Fig. 12.4). Aldosterone (Aldo) triggers nuclear translocation of cytoplasmic MR, binding as a dimer to HREs, and stimulation of transcription initiation complex formation (arrow, upstream of the so-called protein-coding gene, defines the site of transcription initiation). See text for further details.

Fig_12_3

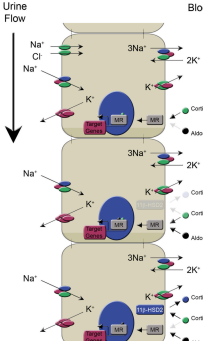


Fig. 12.6 Schematic of principal cells in the aldosterone-sensitive distal nephron (ASDN). The ASDN includes the distal third of the distal convoluted tubule (DCT), connecting tubule, and collecting duct. The Na⁺/K⁺-ATPase (NKA) is located on the basolateral membrane of the cells, and the Na⁺ channel (NCH) is located on the apical membrane. Aldosterone binds to MR on the intracellular side of the Na⁺ channel, leading to increased Na⁺ reabsorption. The diagram also shows the effect of Aldosterone on the Na⁺ ATPase, leading to increased Na⁺ reabsorption.

Fig_12_6

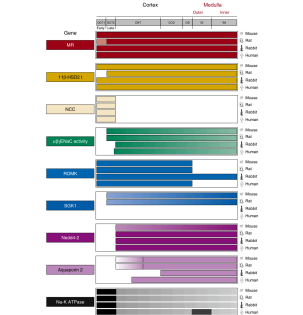


Fig. 12.9 Expression and/or activity of the mineralocorticoid-dependent transport receptors in principal cells along the rat distal nephron. The figure shows a bar chart of the expression and/or activity of various transporters (ENaC, NCC, NKCC, ROMK, HCN, GIRK, PDK1, PDK2, PDK3, PDK4, PDK5, PDK6, PDK7, PDK8, PDK9, PDK10, PDK11, PDK12, PDK13, PDK14, PDK15, PDK16, PDK17, PDK18, PDK19, PDK20, PDK21, PDK22, PDK23, PDK24, PDK25, PDK26, PDK27, PDK28, PDK29, PDK30, PDK31, PDK32, PDK33, PDK34, PDK35, PDK36, PDK37, PDK38, PDK39, PDK40, PDK41, PDK42, PDK43, PDK44, PDK45, PDK46, PDK47, PDK48, PDK49, PDK50, PDK51, PDK52, PDK53, PDK54, PDK55, PDK56, PDK57, PDK58, PDK59, PDK60, PDK61, PDK62, PDK63, PDK64, PDK65, PDK66, PDK67, PDK68, PDK69, PDK70, PDK71, PDK72, PDK73, PDK74, PDK75, PDK76, PDK77, PDK78, PDK79, PDK80, PDK81, PDK82, PDK83, PDK84, PDK85, PDK86, PDK87, PDK88, PDK89, PDK90, PDK91, PDK92, PDK93, PDK94, PDK95, PDK96, PDK97, PDK98, PDK99, PDK100) in the Cortex and Medulla. The expression and/or activity is shown as a bar chart, with the height of the bar indicating the level of expression and/or activity. The data is summarized in the following table.

Fig_12_8

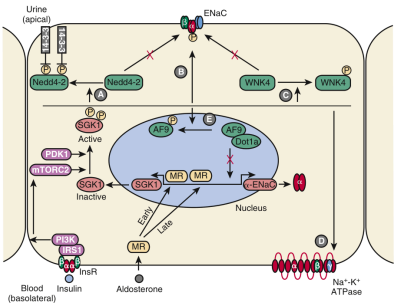


Fig. 12.9 Mechanisms of serum- and glucocorticoid-regulated kinase 1 (SGK1)-mediated stimulation of the amiloride-sensitive sodium channel (ENaC). Within principal cells of the mammalian kidney, SGK1 is transcriptionally upregulated as an early aldosterone-induced gene product. SGK1 is then phosphorylated twice via a phosphatidylinositol-3-kinase (PI3K)-dependent cascade of upstream kinases leading to active SGK1. Active SGK1 has multiple effects: it increases apical plasma membrane ENaC by inhibiting Nedd4-2 and Pdk1¹, and it induces transcription of the α -ENaC (thereby influencing late effects of aldosterone). (A-E, clockwise) Shown are the individual mechanisms that have been elucidated in principal cells. See text for details. *InsR*, insulin receptor; *IRS1*, insulin receptor substrate 1; *MR*, mineralocorticoid receptor; *mTORC2*, mammalian target of rapamycin complex 2; *PDK1*, 3-phosphoinositide-dependent protein kinase type 1.

Fig_12_9